



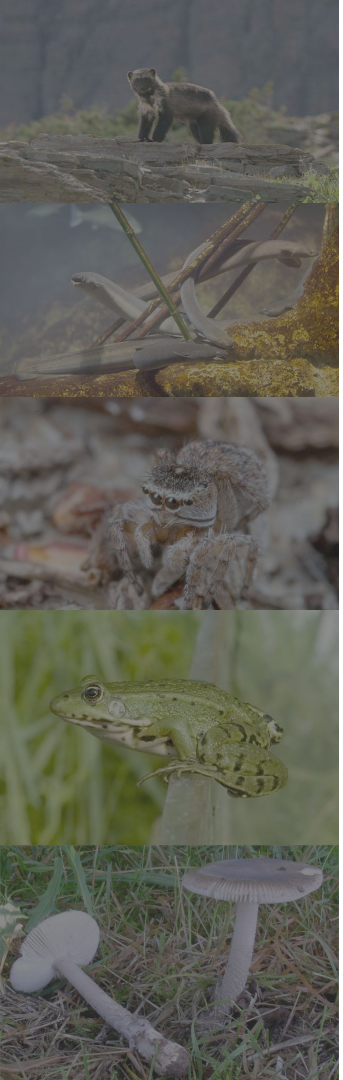




Sampling Bias

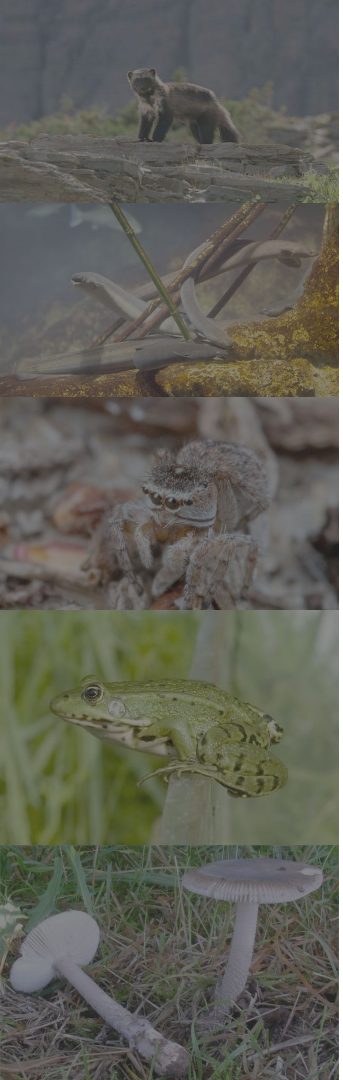
Understanding data collection

Sam Wenaas Perrin



All models are wrong, some are useful

George Box

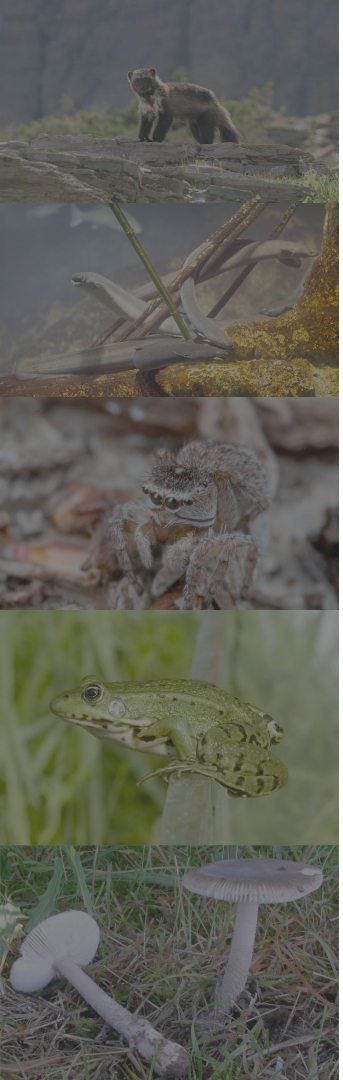


3,846,589,630
Occurrence records

124,809
Datasets

2,817
Publishing institutions

16,018
Peer-reviewed papers using data



3,846,589,630
Occurrence records

124,809
Datasets

2,817
Publishing institutions

16,018
Peer-reviewed papers using data

3,516,241,759
Occurrence records

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Datasets





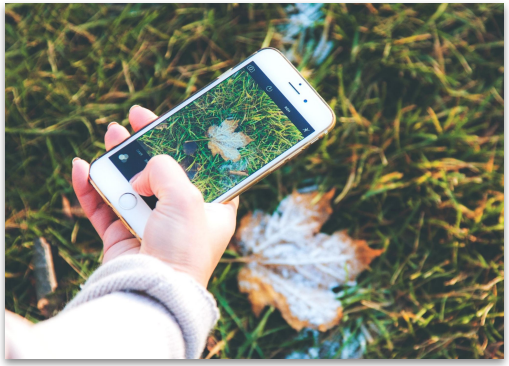


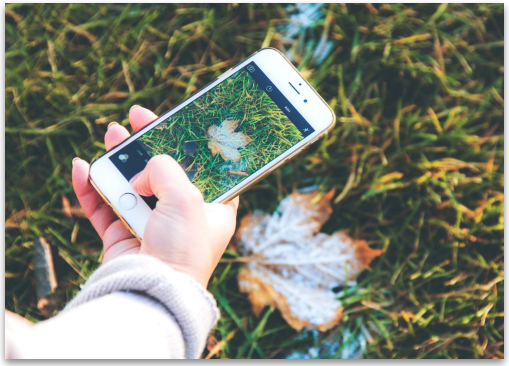


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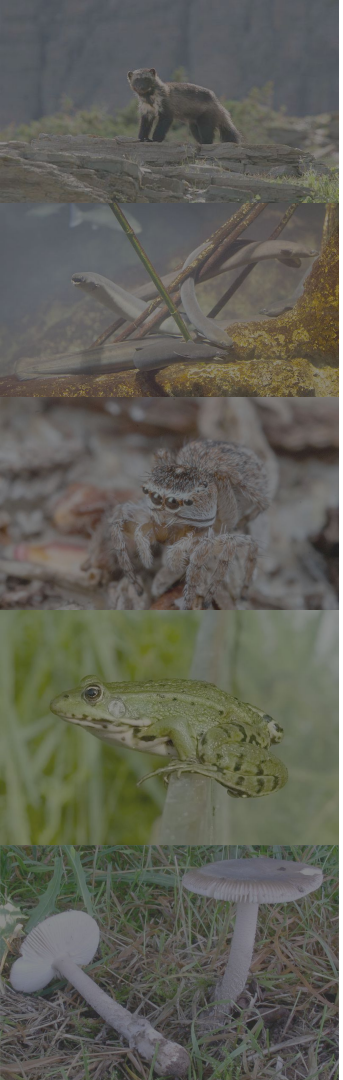





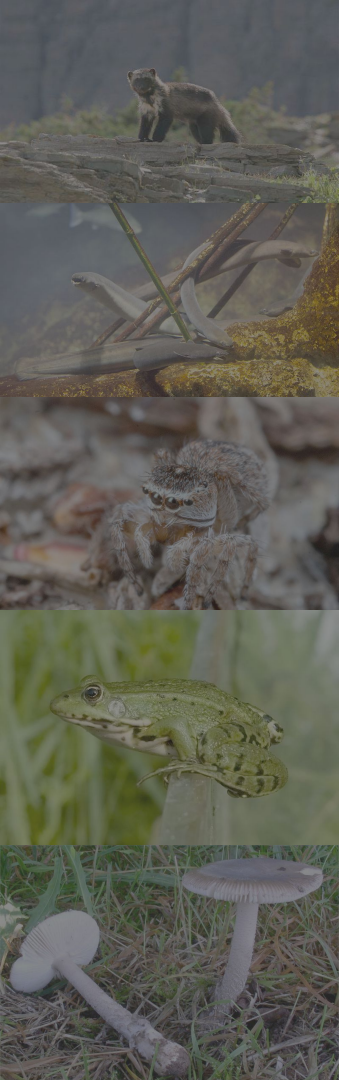

iNaturalist

Artsorakelet kjenner igjen bilder av arter





Spatial Bias



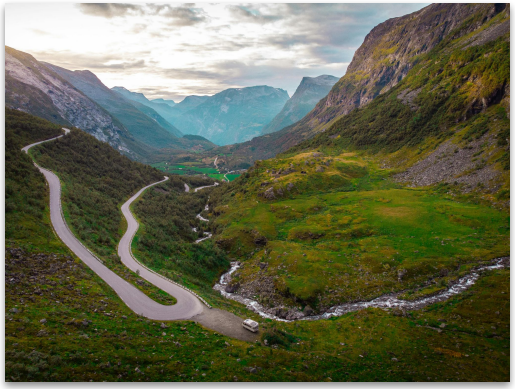
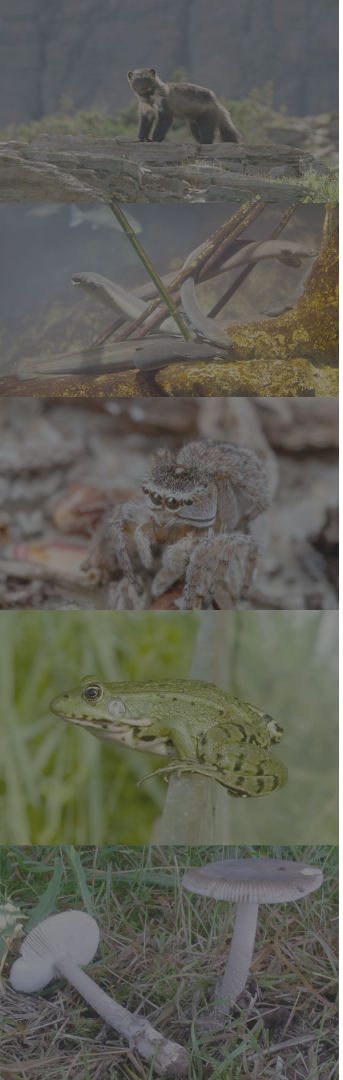


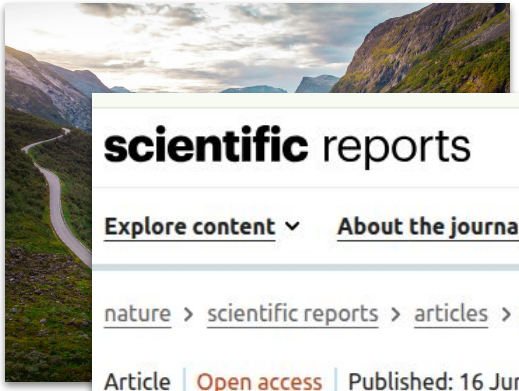


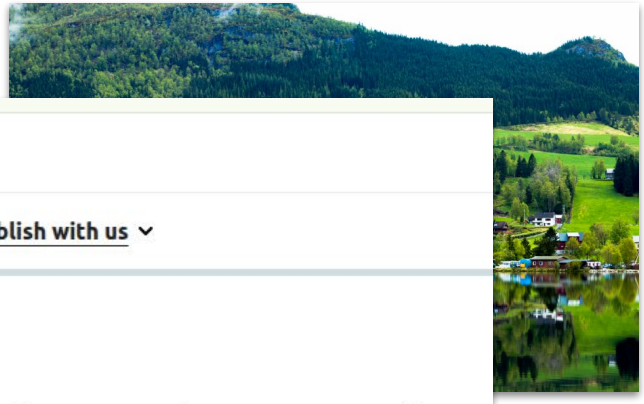




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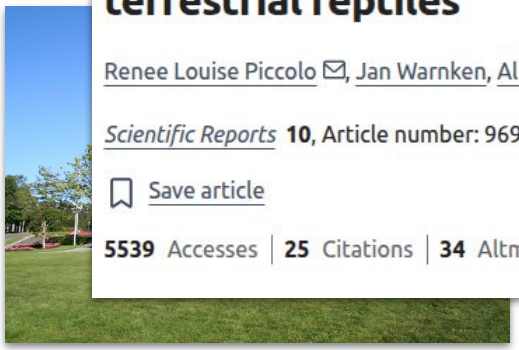
Location biases in ecological research on Australian terrestrial reptiles

Renee Louise Piccolo, Jan Warnken, Alienor Louise Marie Chauvenet & James Guy Castley

Scientific Reports **10**, Article number: 9691 (2020) | Cite this article

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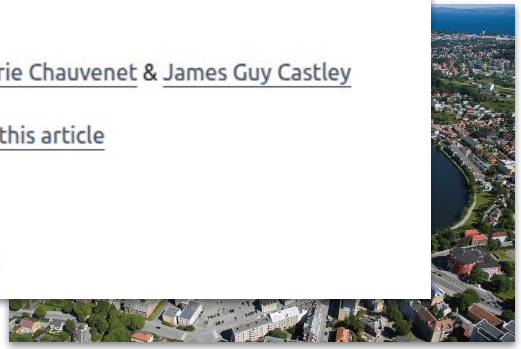
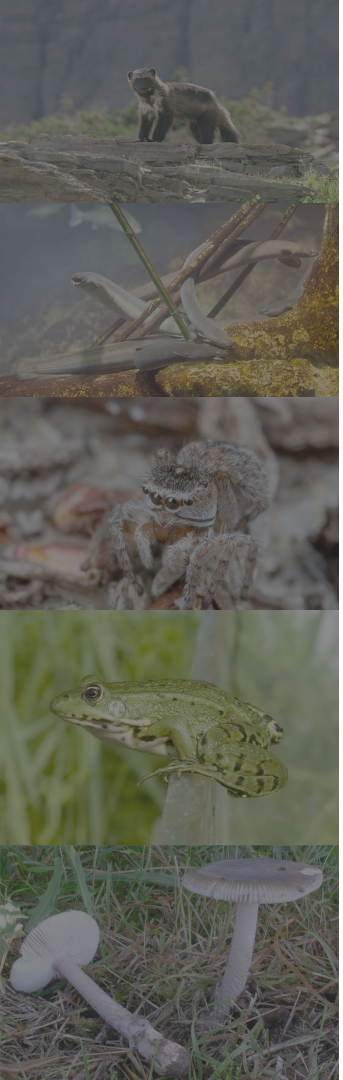


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Temporal Bias

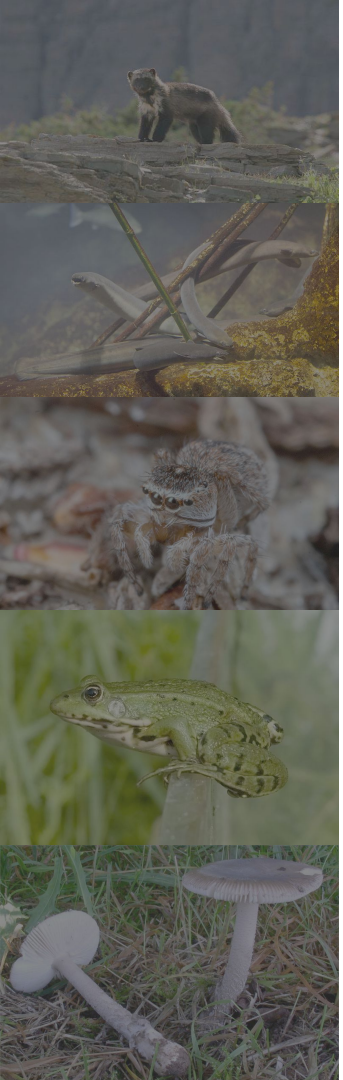










Image credit: Linerle - Bingar1234 (CC BY-SA 4.0)

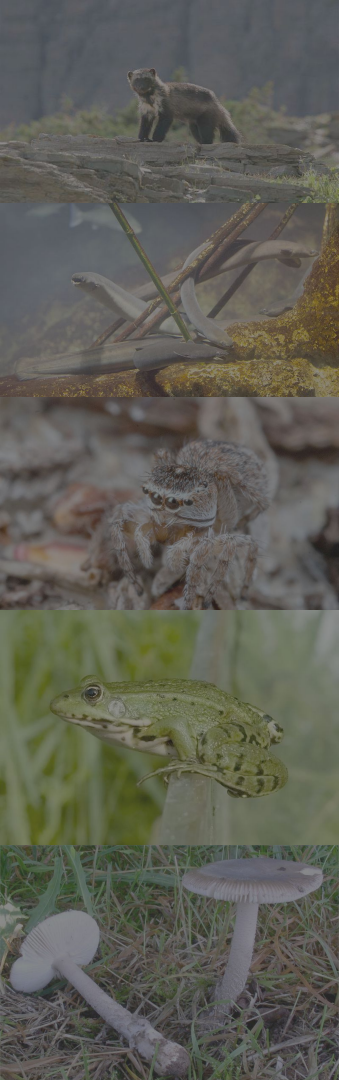




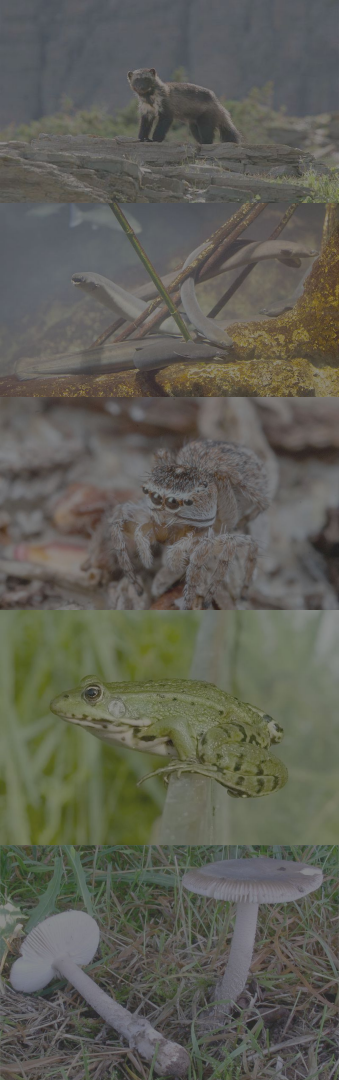








Image credit: Moose - Animalia (CC BY-SA 3.0)





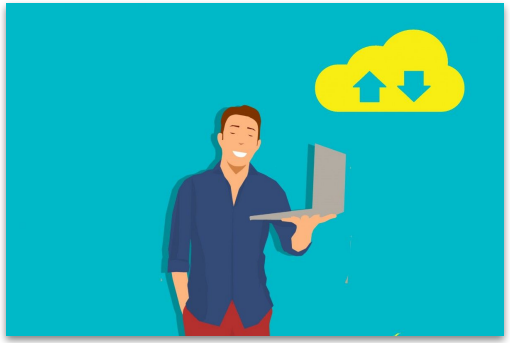






Image credit: Pink salmon - Earl Steele (CC BY-NC-ND 2.0)

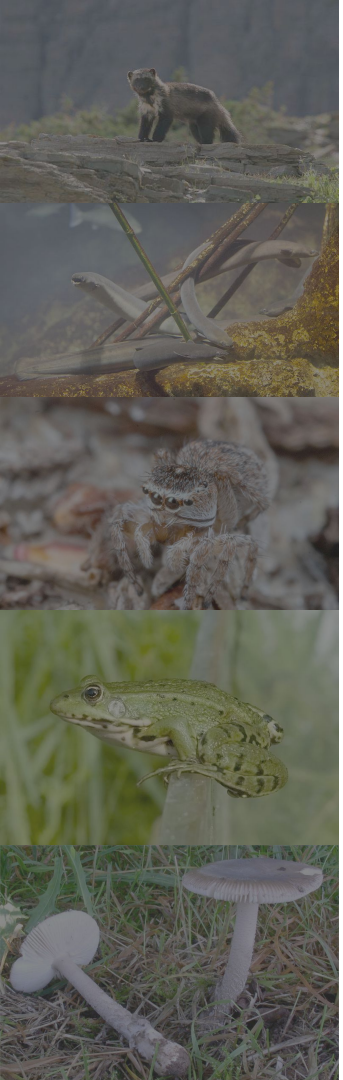
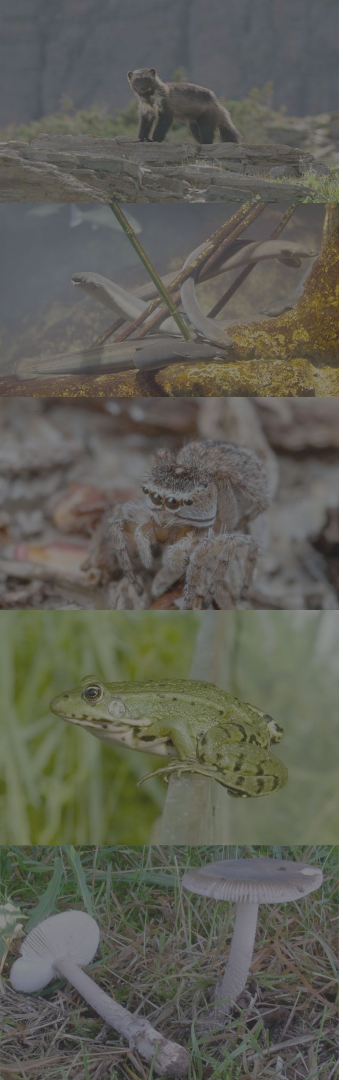
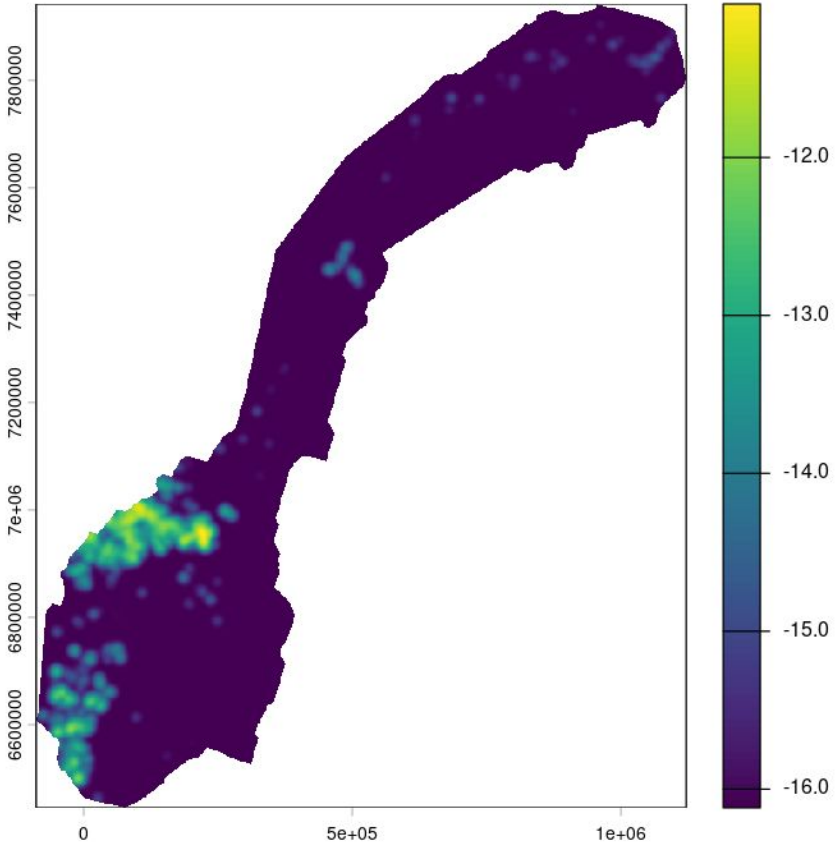




Image credit: Daniel Neale (CC BY 2.0)







John Bjarne Jordal

Virksomhet | Aktuelle prosjekter | Publikasjoner | Glimt fra artsmangfoldet | Kontakt

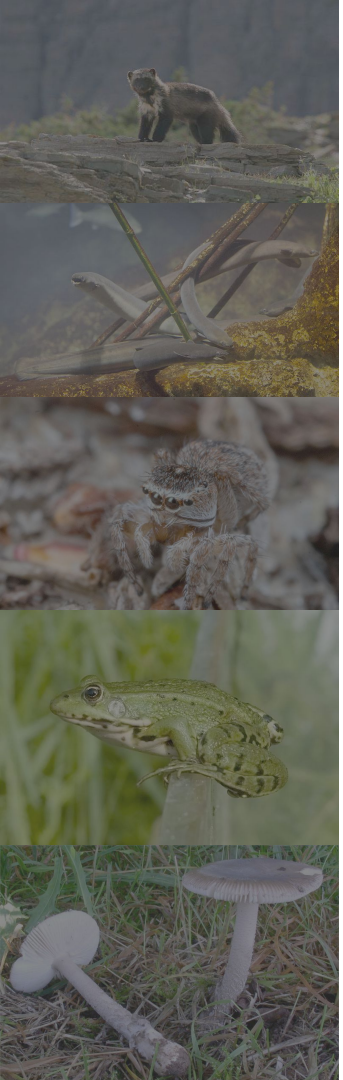
Bakgrunn

- Cand. real. fra NTNU (1979), fagkrets biologi, kjemi, matematikk, pedagogisk kompetanse i de samme fag, hovedfag i terrestrisk zoologi (feromoner hos barkbiller på furu), agronomeksamen (1982), en rekke sommerjobber med feltregistreringer 1974-1980 (Sør-Norge, Svalbard, Nord-Norge)
- Lektor ved Fræna videregående skole 10 år (1979-1989) med biologi og naturfag, kjemi og matematikk som undervisningsfag, pluss div. funksjoner som verneombud, samlingsstyrer mm.
- Ressurssenterdrift ved Tingvoll videregående skole hele 1990-tallet, kurs om miljøspørsmål for kommuner og næringsliv, prosjekter for Utdannings- og forskningsdepartementet, kursmoduler på universitets- og høyskolenivå.
- Enkeltpersonsforetak i biologi og naturforvaltning 1992-2012.
- Tilsatt i eget firma (Biolog J.B. Jordal) 2013-2021.
- Tilsatt i Miljøfaglig Utredning (www.mfu.no) fra 01.01.2019.
- Rundt 390 publikasjoner, en rekke kurs, foredrag mm.

Virksomhet – biologi og naturforvaltning (jf. https://www.mfu.no/)

- Utredninger og utviklingsarbeid vedrørende biologisk mangfold, truete arter og bevaringsbiologi, spisskompetanse på mangfold i kulturlandskapet
- Kartlegging av prioriterte naturtyper etter DN-håndbok 13 og Miljødirektoratets instruks (NiN-basert kartlegging)
- Oppbygging av databaser
- Forvaltnings- og skjøtselsplaner, særlig tilknyttet verdifulle kulturlandskap
- Konsekvensutredninger i forbindelse med planlagte tiltak
- Kurs, foredrag, markdager mm.

Se også https://www.mfu.no/john-bjarne-jordal

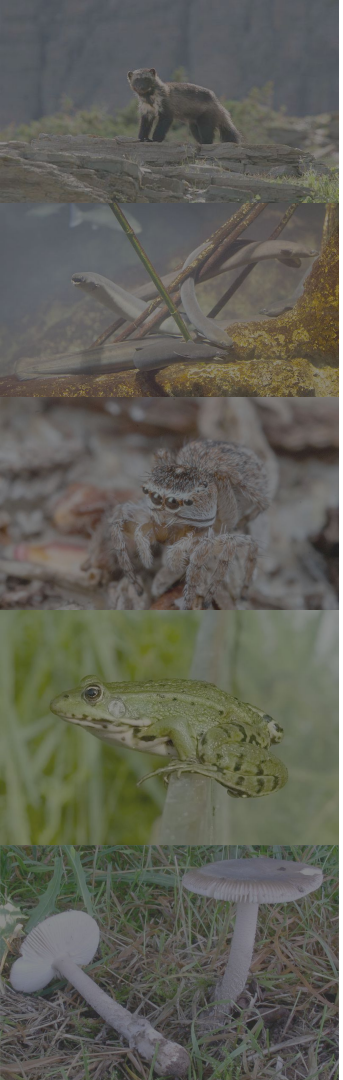


Species Bias







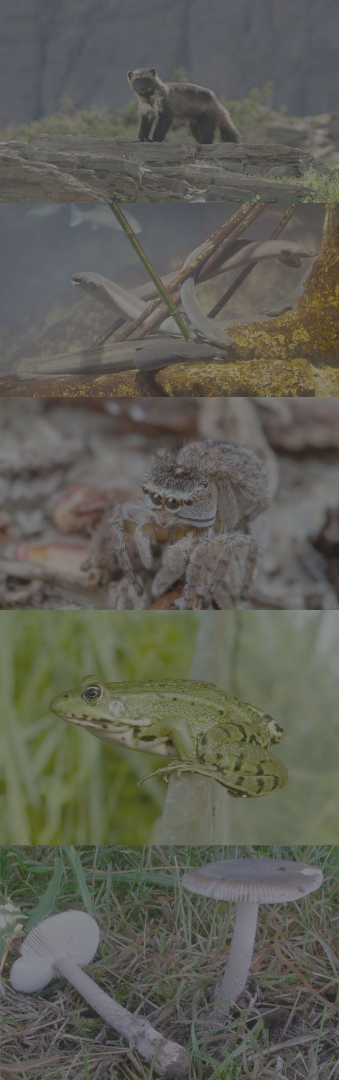


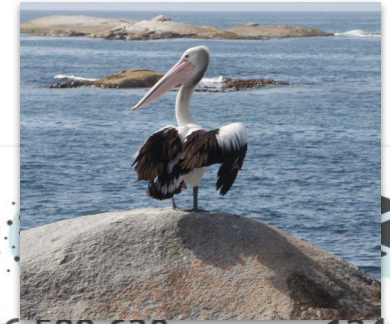
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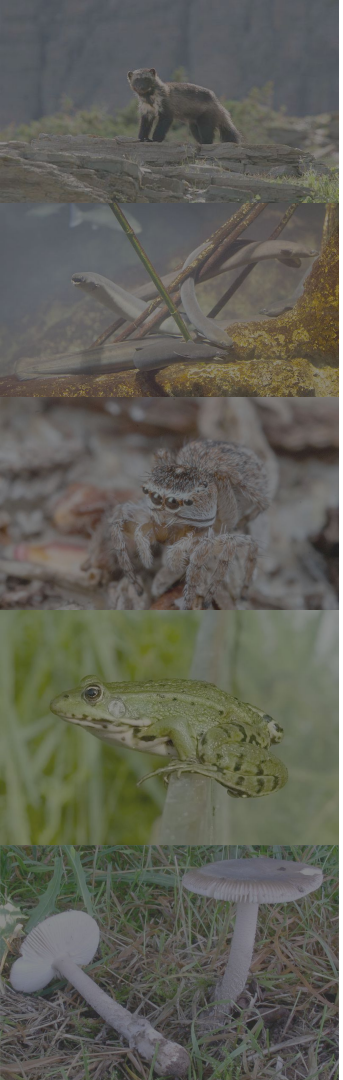






16,018

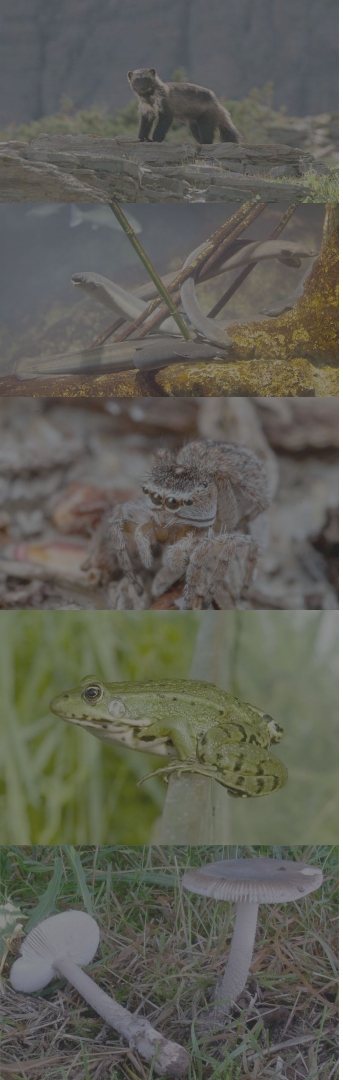
Peer-reviewed papers using data



Some species are harder to see

Some species are harder to identify

Some species are just less interesting



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Opinion

We need to talk about nonprobability samples

Robin J. Boyd [1] @, Gary D. Powney [1] @, Oliver L. Pescott [1] @ Show more

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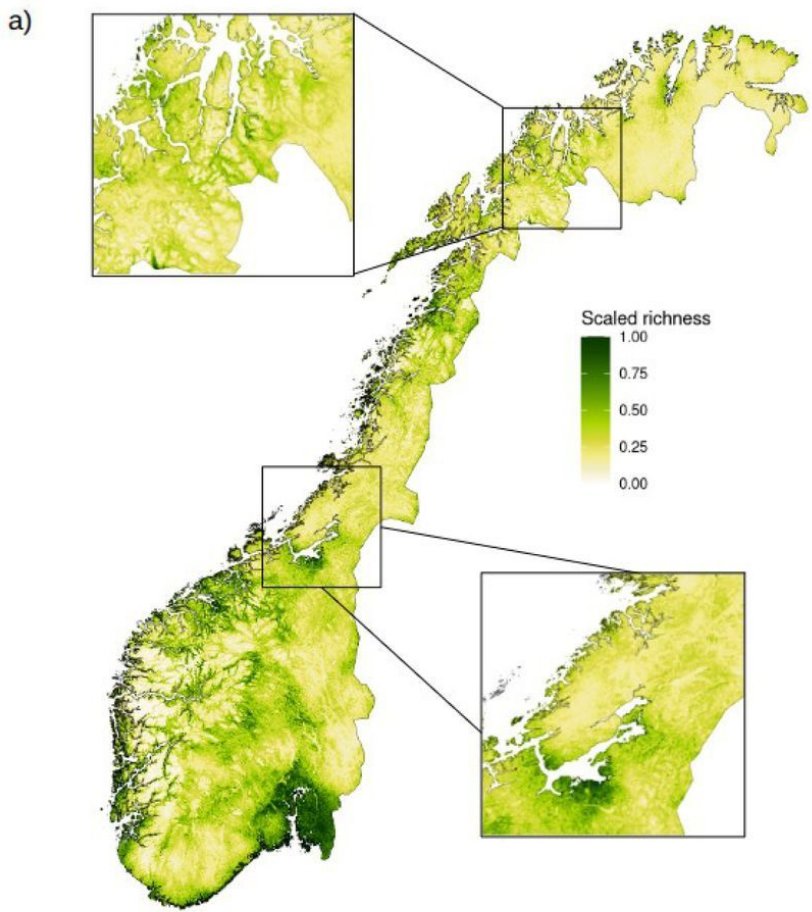


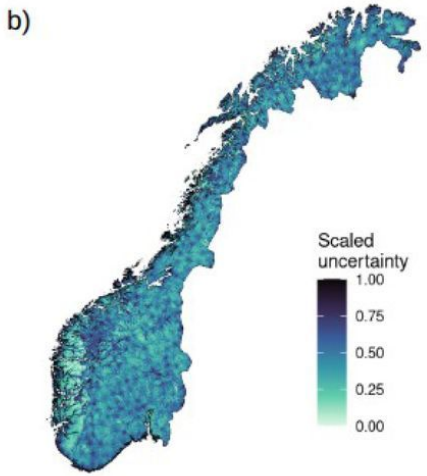


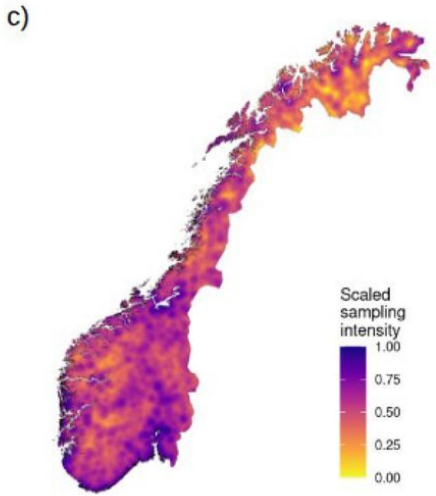


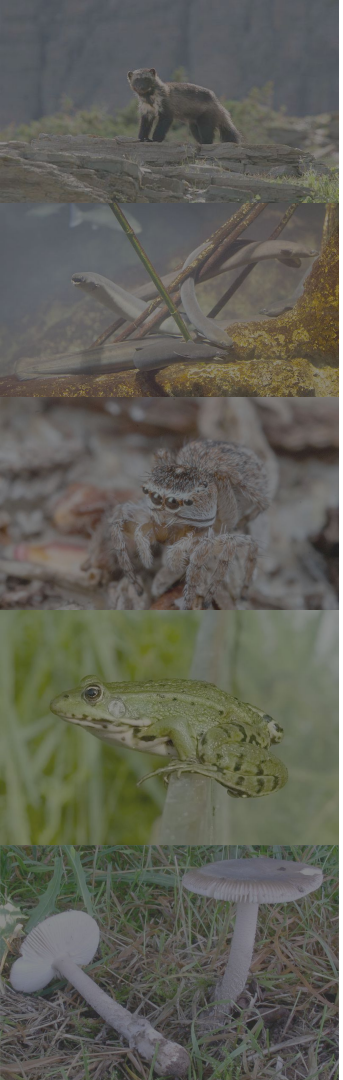




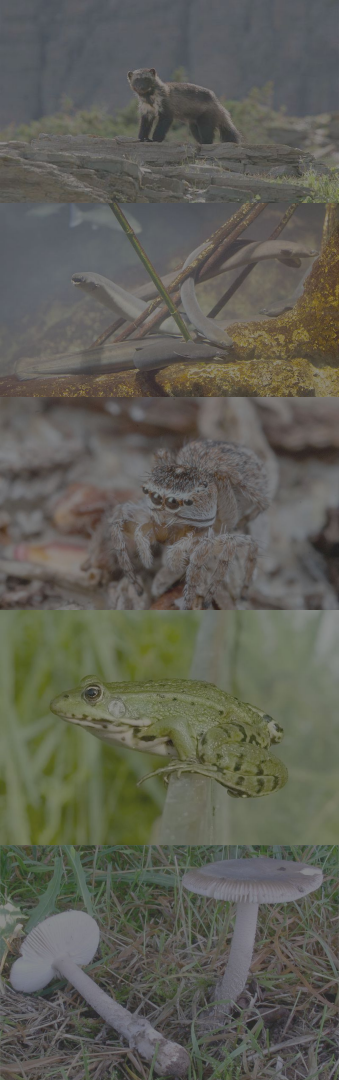

a)
Scaled richness
1.00
0.75
0.50
0.25
0.00


b)
Scaled uncertainty
1.00
0.75
0.50
0.25
0.00


c)
Scaled sampling intensity
1.00
0.75
0.50
0.25
0.00



So if we know about all these biases, can’t we just account for them?



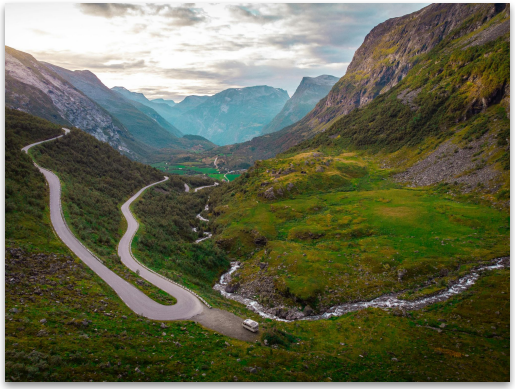
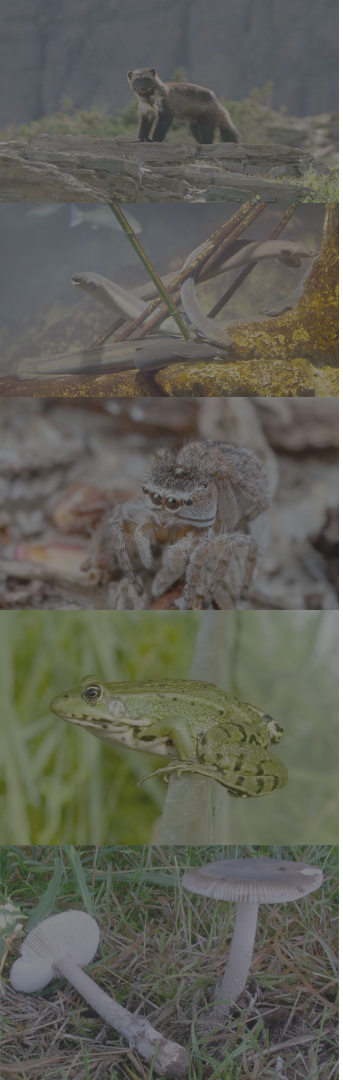


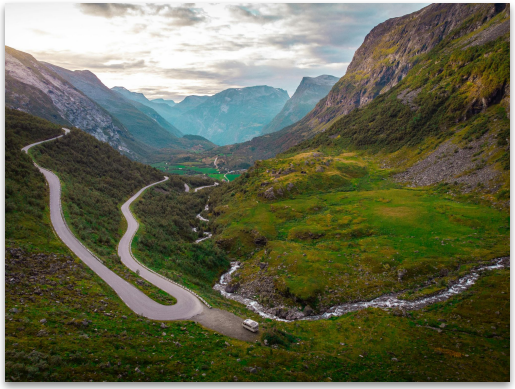






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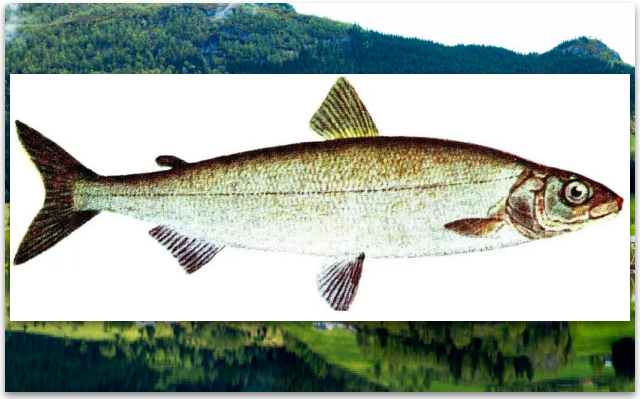






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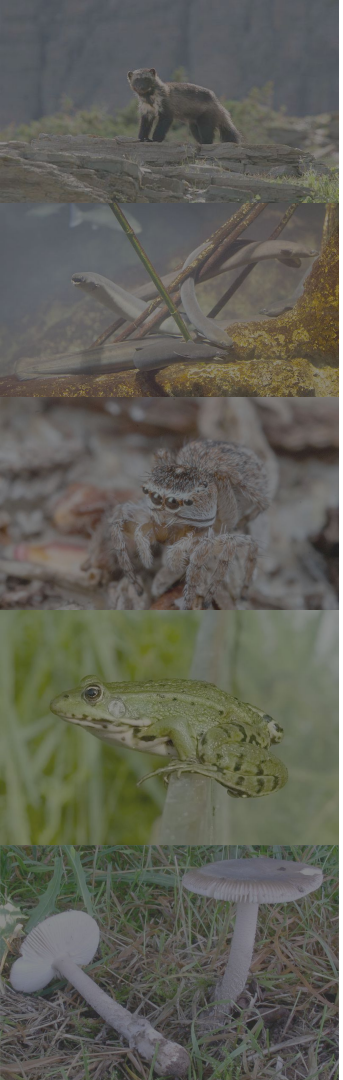


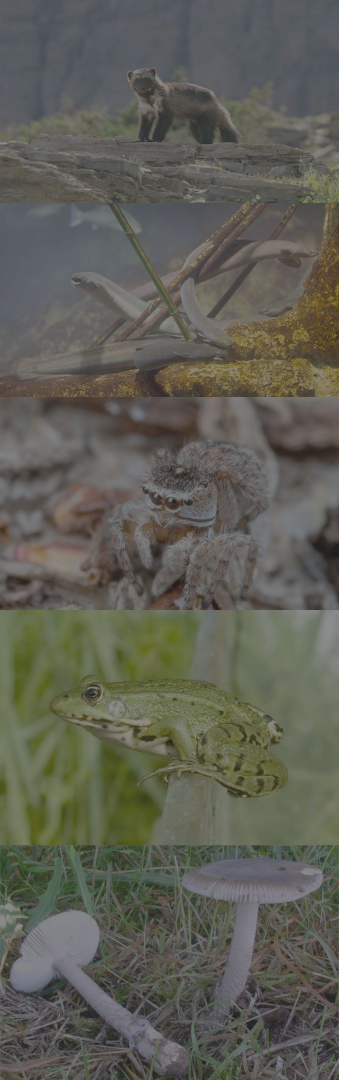




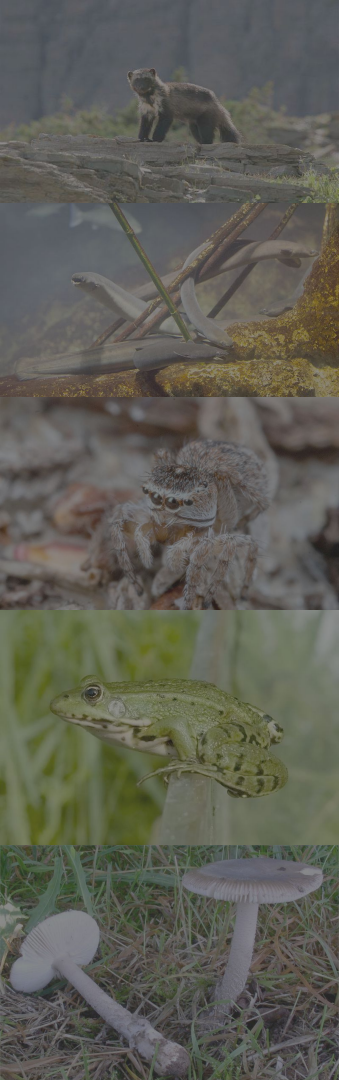




Image credit: Moose - Animalia (CC BY-SA 3.0)



Does representative sampling exist?



Sampling event | Registered 25 October 2024

Swedish National Forest Inventory: Presence-absence vegetation data (1995-current)

Published by Swedish National Forest Inventory, Department of Forest Resource Management, SLU

Dahlgren J www.slu.se Licence CC0 1.0 43 citations 8,817,270 occurrences

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Description

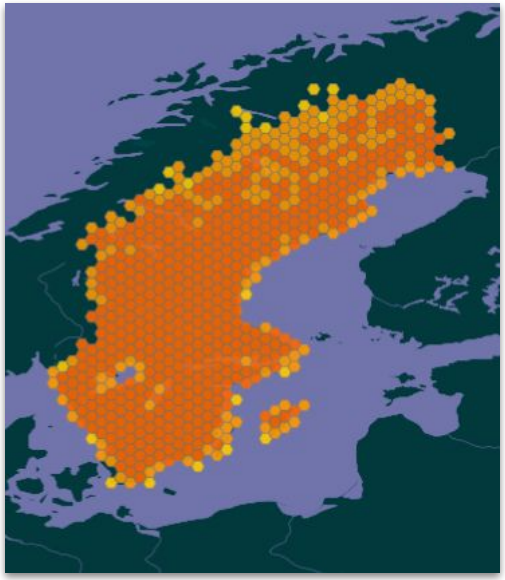
Publication date: 12 February 2026

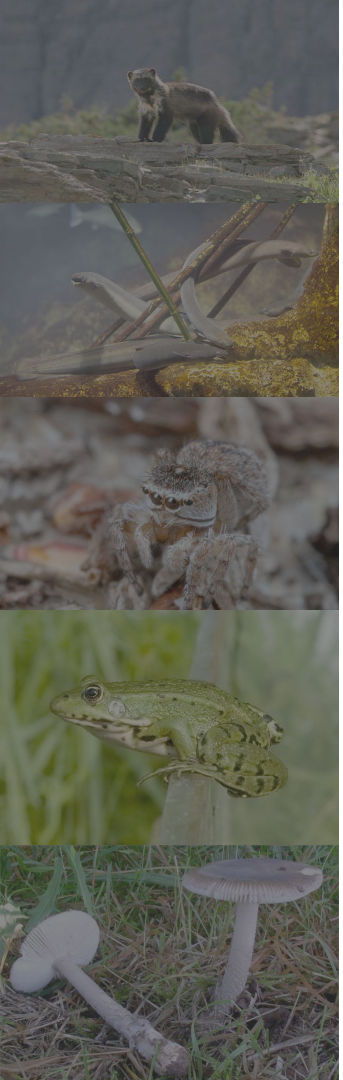
The Swedish National Forest Inventory (NFI) is a sample based field inventory performed by the Swedish University of Agricultural Sciences (SLU). The annual inventory is undertaken on sample plots and this forms the basis for a large number of estimates. Data is collected across the whole of Sweden and statistics are published for all land use classes except urban land and sea and fresh water.

The data collection is focused on forest land and particularly productive forest land. Since year 2003 areas within formally protected land are included and since year 2016 non–coniferous alpine forest are included.

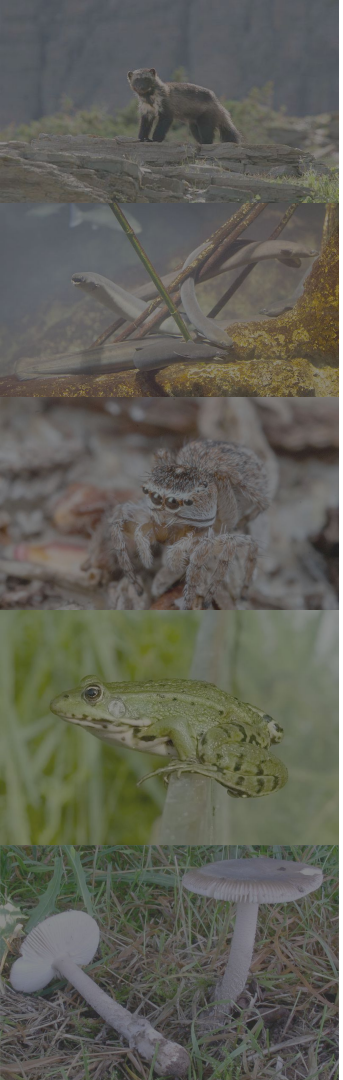
The sample size (number of permanent sample plots where presence/absence registrations are performed) was reduced by 50% from year 2018 and onwards. This dataset consists of the reduced sample. The supplement data from the discontinued sample for the period 1995-2017 is available in the related dataset 'Swedish National Forest Inventory: Presence-absence vegetation data (1995-2017, supplement sample)'.

On this page
- Description
- Purpose
- Geographic scope
- Temporal scope
- Taxonomic scope
- Methodology
- Additional info
- Bibliography
- Contacts
- GBIF registration
- Citation

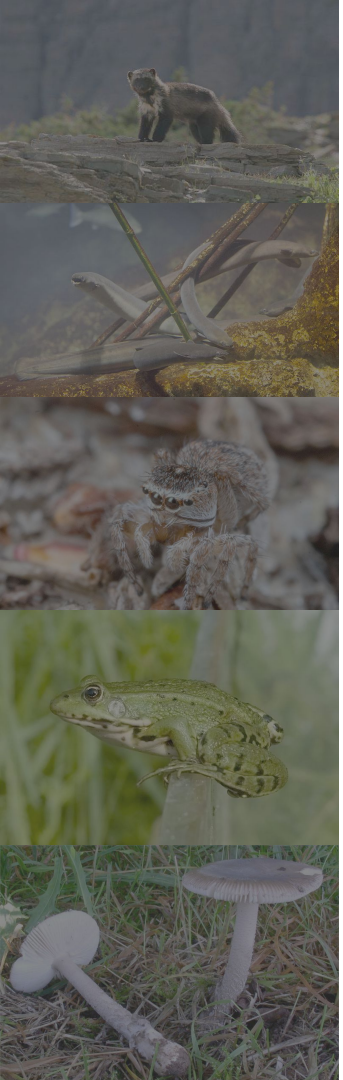




Human inference is always going to be necessary



Human inference is always going to be necessary



Questions to ask

- What scale are you looking at?
- What are the nearby drivers of sampling bias?
- Do you benefit from using biased data?
- How tightly connected are sampling bias and species richness?
- Does sampling bias matter for your final output?